DataWise — Usage Guide

How to run DataWise and use every major feature (tabular ML cleaning, documents, images, audio, video, archives). Written so anyone can follow the steps and re-run the same tests.

A shareable PDF copy of this guide is available as `USAGE_GUIDE.pdf` in the repo root (regenerate with `python scripts/export_usage_guide_pdf.py`).

1. Start the app

From the repo root (`DataWise/`):

```
# One-time (if not done): create venv + install
python -m venv .venv
.venv\Scripts\Activate.ps1
pip install -r requirements.txt

# Every day — easiest
.\start.ps1
```

Or two terminals:

```
# Terminal 1 — API
.venv\Scripts\Activate.ps1
python -m uvicorn backend.main:app --host 127.0.0.1 --port 8000 --workers 1

# Terminal 2 — UI
.venv\Scripts\Activate.ps1
streamlit run frontend/app.py --server.port 8501
```

Open:

Service	URL
UI	http://localhost:8501
API health	http://127.0.0.1:8000/health
API docs	http://127.0.0.1:8000/docs

Important: always use `--workers 1` for the API.

2. General workflow (all file types)

- Open the UI.
- Upload one (or more) files.
- Type a plain-English prompt.
- Click **Run Pipeline**.
- Review preview / summary / plan steps.
- Download the result.

Same idea via API:

```
curl -X POST "http://127.0.0.1:8000/api/pipeline/run" `
  -F "prompt=clean this for ML" `
  -F "file=@test_files/tabular/messy_customers.csv"
```

3. Tabular files (CSV / JSON / Excel) — ML cleaning

Test files

File	Path
Messy CSV	`test_files/tabular/messy_customers.csv`
Messy JSON	`test_files/tabular/messy_customers.json`
Messy Excel	`test_files/tabular/messy_customers.xlsx`

Also see older samples under `test_data/` and `tests/fixtures/sample.csv`.

What `clean_for_ml` does

When you ask to clean tabular data, DataWise:

- Strips whitespace
- Turns `N/A` / `null` / blanks / `-` / `?` into missing values
- Drops empty columns
- Removes duplicate rows
- Coerces numeric text → numbers
- Imputes remaining nulls (median / mode)
- Standardizes dates to `YYYY-MM-DD`
- Drops constant (no-signal) columns

Prompts to try

Prompt	Expected
`clean this for ML`	Full ML prep pipeline
`clean this CSV so I can use it in my ML project`	Same
`prepare this data for machine learning`	Same
`sort by Age ascending`	Sort only (all rows)
`show only Name and Age columns`	Column projection
`only display 2 columns`	Keep first 2 columns
`filter rows where Age > 30`	Row filter

Prompt	Expected
`group by Region and sum Purchase`	Groupby aggregate
`top 5 by Purchase`	Sort + top N
`remove the Phone column`	Drop column (if present)

How to test (manual)

- Upload `test_files/tabular/messy_customers.csv`.
- Prompt: `clean this for ML`.
- Confirm rows drop (dupes gone), empty/constant columns removed, download works.
- Repeat with `.json` and `.xlsx`.

How to test (API smoke)

```
.venv\Scripts\Activate.ps1
python -c "import httpx; from pathlib import Path; p=Path('test_files/tabular/messy_customers.csv'); r=httpx.post(
```

4. Documents (PDF / DOCX / TXT / MD)

Test files

Generate (or refresh) fixtures:

```
.venv\Scripts\python.exe scripts\generate_document_test_files.py
```

File	Path	Best for
PDF report	`test_files/documents/sample_report.pdf`	text, entities, pages, links, metadata
DOCX invoice	`test_files/documents/sample_invoice.docx`	tables→CSV, headings, key-values
TXT report	`test_files/documents/sample_report.txt`	entities, redact, summary
Markdown	`test_files/documents/sample_overview.md`	headings, links

Features and prompts

Feature	Example prompt	Notes
Extract text	`extract text from this document`	Default baseline
Clean text	`clean this document text`	Whitespace + header/footer noise

Feature	Example prompt	Notes
Word count	`count the words in this document`	
Metadata	`extract document metadata`	pages, author, size, etc.
Headings / outline	`extract headings` / `show the outline`	PDF/DOCX/MD
Tables → CSV	`extract all tables`	Writes `table_1.csv`, … in session folder
Split by page	`split this PDF by page`	Writes `page_1.txt`, …
Entities	`extract emails, phones, dates, and amounts`	Deterministic regex
Key-value / invoice fields	`extract invoice fields` / `extract key values`	`Label: value` lines
Find mentions	`find all mentions of "Revenue"`	Quotes recommended
Links	`extract all links and urls`	URLs + emails
Language	`detect the language of this document`	Lightweight heuristic
Images / figures	`extract images from this PDF`	PDF only
Redact PII	`redact emails and phone numbers`	Masks PII in text
Summary	`summarize this document`	Extractive (no LLM required)
Q&A	`what is the invoice number?`	Uses LLM if API key set; else returns extracted text fallback
OCR (scanned PDF / image)	`OCR this scanned PDF` / `extract text with OCR`	Needs Tesseract — see **§5 OCR setup**
Convert / export	`convert this document to markdown`	Download as txt/md/json/docx/pdf

How to test each document feature (checklist)

- Start backend + UI.
- Upload `test_files/documents/sample_invoice.docx`.
- Run: `extract all tables` → download / check session for CSV artifacts.
- Upload `sample_report.pdf`.
- Run in turn:
- `extract text from this document`
- `extract emails, phones, dates, and amounts`

- `extract invoice fields`
- `find all mentions of "Revenue"`
- `split this PDF by page`
- `extract document metadata`
- `summarize this document`
- Upload `sample_report.txt` → `redact emails and phone numbers`.
- Upload `sample_overview.md` → `extract headings`.
- After OCR is installed (§5): upload a scanned PDF / image → `OCR this scanned PDF` or `extract text with OCR`.

5. OCR setup (Tesseract) — share these steps with friends

OCR lets DataWise read **scanned PDFs** (image-only pages) and **extract text from images**. Normal text-layer PDFs / DOCX / TXT work **without** OCR. Install this only if you need scanned-doc or image text extraction.

What gets installed

Piece	Role
Tesseract OCR (system app)	The actual engine that reads text from images
pytesseract (Python package)	Bridge between DataWise and Tesseract

DataWise looks for Tesseract in this order: `TESSERACT_BINARY` in `.env` → PATH → common Windows paths like `C:\Program Files\Tesseract-OCR\tesseract.exe`.

Step A — Install the Tesseract system binary

Windows (recommended)

Open **PowerShell** and run:

```
winget install --id UB-Mannheim.TesseractOCR -e --source winget --accept-package-agreements --accept-source-agreem
```

Then **close and reopen** the terminal (so PATH refreshes), and verify:

```
tesseract --version
```

You should see a version line (e.g. `tesseract 5.x.x`).

If `tesseract` is not found but it installed under Program Files, either:

- Add `C:\Program Files\Tesseract-OCR` to your user PATH, **or**
- Put this in your project `.env` (copy from `.env.example` if needed):

```
TESSERACT_BINARY=C:\Program Files\Tesseract-OCR\tesseract.exe
```

Linux (Ubuntu / Debian)

```
sudo apt update
```

```
sudo apt install -y tesseract-ocr
tesseract --version
```

macOS

```
brew install tesseract
tesseract --version
```

Step B — Install the Python package (inside the DataWise venv)

From the repo root (`DataWise/`):

Windows

```
.venv\Scripts\Activate.ps1
pip install pytesseract
# Or install everything again:
pip install -r requirements.txt
```

Linux / macOS

```
source .venv/bin/activate
pip install pytesseract
# Or:
pip install -r requirements.txt
```

`pytesseract` is already listed in `requirements.txt`, so a normal `pip install -r requirements.txt` is enough if the venv is fresh.

Step C — Verify DataWise can see Tesseract

With the venv active, from the repo root:

```
python -c "from backend.core.ocr_locator import get_tesseract_path, is_tesseract_available; print('available:', is
```

Expected:

```
available: True
path: C:\Program Files\Tesseract-OCR\tesseract.exe
```

(Path may differ on Linux/macOS.)

If `available: False`, set `TESSERACT_BINARY` in `.env` to the full path of `tesseract` / `tesseract.exe`, save, and run the check again.

Step D — Restart DataWise

Stop the backend and UI, then start again (so they pick up env / PATH):

```
.\start.ps1
```

Or use **Run and Debug** → **DataWise: Full Stack** (F5).

Step E — Test OCR in the UI

- Open http://localhost:8501

- Try one of these:

File to upload	Prompt
Scanned / image-only PDF	`OCR this scanned PDF`
Scanned PDF	`extract text` (auto-OCR if there is no text layer)
JPG / PNG image	`extract text with OCR`

- Confirm the result text appears (and download works).

Fixture tip: `tests/fixtures/documents/sample_scanned.pdf` is an image-only PDF made for OCR testing.

OCR prompts (cheat sheet)

Prompt	Works on
`OCR this scanned PDF`	PDF
`extract text with OCR`	PDF or image
`extract text`	PDF — uses OCR automatically if no text layer

OCR troubleshooting

Problem	Fix
`tesseract: command not found`	Reopen terminal after install; or set `TESSERACT_BINARY` in `.env`
`available: False` in Step C	Point `.env` at the real `tesseract.exe` path; restart backend
`ModuleNotFoundError: pytesseract`	Activate `.venv` and run `pip install pytesseract`
OCR returns empty / garbage text	Scan quality is low; try a clearer image (Tesseract limitation)
Text PDFs work, scanned ones fail	OCR binary missing — finish Steps A–D
Friend on another PC	They must run Steps A–E on **their** machine (Tesseract is not bundled in the repo)

6. Images

Supported ops (see also `EXAMPLE_COMMANDS.md`):

Prompt example	Result
`resize this image to 1024x1024`	Resize
`crop image to 50, 50, 450, 350`	Crop

Prompt example	Result
`rotate this image by 90 degrees`	Rotate
`convert image to PNG`	Format convert
`compress this image with quality 50`	Compress
`extract image metadata and dimensions`	Metadata
`extract text with OCR`	Read text from the image (needs §5)

Fixtures: `tests/fixtures/images/` (generate via `scripts/generate_image_fixtures.py` if needed).

7. Audio

Prompt example	Result
`extract audio metadata`	Duration, sample rate, channels
`trim this audio to first 2 seconds`	Trim
`convert audio to mp3`	Convert
`compress this audio`	Compress

Requires **FFmpeg** on PATH (`ffmpeg -version`).

8. Video

Prompt example	Result
`extract video metadata`	Metadata
`trim this video to first 3 seconds`	Trim
`extract frame at 1 second`	Frame image
`extract audio from this video`	Audio track
`resize this video`	Resize (when planned)

Requires **FFmpeg**.

9. Archives (ZIP / TAR / GZ)

Prompt example	Result
`list files in this archive`	Member listing
`extract and process all files in archive`	Safe extract + process

Security: path traversal / zip-bomb guards are built in.

10. Automated tests (developers)

```
.venv\Scripts\Activate.ps1

# Full suite
.venv\Scripts\pytest.exe -q --tb=short

# Useful subsets
.venv\Scripts\pytest.exe backend/tests/test_tools_clean.py -q
.venv\Scripts\pytest.exe backend/tests/test_document_processor.py -q
.venv\Scripts\pytest.exe backend/tests/test_pipeline.py -q
.venv\Scripts\pytest.exe backend/tests/test_regression_emergency_fix.py -q
```

OCR-specific smoke (needs Tesseract — see §5):

```
.venv\Scripts\pytest.exe backend/tests/test_document_processor.py::test_ocr_scanned_pdf backend/tests/test_image_p
```

11. Feature map (quick reference)

Area	Main capability	Test folder
Tabular ML clean	`clean_for_ml`	`test_files/tabular/`
Documents	extract / tables / entities / redact / summary / …	`test_files/documents/`
OCR	scanned PDF + image text	§5 + `tests/fixtures/documents/sample_scanned.pdf`
Images	resize, crop, rotate, convert, compress, OCR	`tests/fixtures/images/`
Audio / video	metadata, trim, convert (FFmpeg)	`tests/fixtures/audio/`, `tests/fixtures/video/`
Archives	list / safe extract	`tests/fixtures/archives/`

12. Related docs

Doc	Purpose
`PROJECT_SETUP.md`	Install from scratch
`PROJECT_OVERVIEW.md`	Architecture
`EXAMPLE_COMMANDS.md`	More prompt examples
`capabilities.json`	Enabled processors / limits
`test_files/README.md`	Index of tabular + document fixtures
`test_files/documents/README.md`	Document fixtures

Doc	Purpose
`MANUAL_ACTIONS_LOG.md`	External binaries (Tesseract, etc.)

13. Troubleshooting

Problem	Fix
UI cannot reach API	Start uvicorn on port 8000; check `DATAWISE_API_URL` in `.env`
Weird session errors	Restart API with `--workers 1`
Audio/video fails	Install FFmpeg; reopen terminal
Document / image OCR fails	Follow **§5 OCR setup** (Tesseract + `pytesseract`, optional `TESSERACT_BINARY`, restart app)
Clean prompt does nothing unexpected	Use clearer prompts from this guide
Huge upload rejected	Confirm `.streamlit/config.toml` has `maxUploadSize = 10000`